

Supplementary information

Liver extracellular matrix-based nanofiber scaffolds for the culture of primary hepatocytes and drug screening

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Methods

1. Preparation and characterization of decellularized liver extracellular matrix (LEM)

Rat livers were decellularized according to our previous study¹. Briefly, the rat liver was perfused through the portal vein with detergents 1% SDS (Himedia) and 1% Triton-X-100 (Himedia) with 0.1% NH₄ OH through portal vein for 6 hrs. at a flow rate of 5ml/min (Cole Palmer). Finally, the decellularized liver was washed with distilled water overnight. The decellularized liver tissue was fixed with formalin, dehydrated in a series of ethanol and embedded in paraffin and characterized with histopathological staining as previously reported². The decellularized tissues, were mounted on glass slides post-tissue processing, the tissue slides were then permeabilized with 0.1% TritonX-100 for 15 mins and blocked with 1% BSA for 30mins, later the slides were incubated with DAPI (1:2000) for 2 mins followed by PBS washes, mounted with glycerol and imaged using Leica systems confocal microscopy.

2. Characterization of decellularized liver matrix

For analysis of the matrix proteins, the paraffin-embedded decell tissue sections were stained with Masson's trichrome stain (for collagen), Alcian blue staining (for glycosaminoglycans), and Orcein staining (for elastin protein). All the stained sections were examined under a light microscope and the images were obtained in 10 and 20x.

3. Preparation of electrospinning solution

The LEM solution was also prepared by mixing the LEM powder (10% w/v) in 10 ml of HFIP. Prior to electrospinning, the viscosity of the PLA-LEM solution was measured using a rheometer (Anton Par MCR 302). Electrospinning solutions were prepared by mixing the synthetic polymer solution and LEM solution in a ratio (1:1) and were kept at 4°C under constant stirring overnight.

The solution was then loaded into a 5ml syringe and the electrospinning parameters used to fabricate the nanofiber scaffolds are provided in supplementary table 1. The scaffolds were collected on 13mm glass coverslips in a random orientation which were further dried overnight in a vacuum desiccator.

4. Physical characterization of the electrospun scaffolds

- **Scanning Electron Microscopy (SEM)**

The fiber formation pattern and the respective diameter of the different combinations of electrospun samples (mentioned in Supp Table-2) were assessed using Scanning electron microscopy (SEM). Briefly, each sample was coated with gold sputtering (15-20 nm) using an EMITECH K550X instrument. The representative images were obtained at various magnifications using a JEOL 5610LV (JEOL, Japan) microscope. The respective fiber diameter and pore dimensions were calculated using ImageJ software considering 100 fibers per condition as described previously³. The fiber diameter and pore dimensions are represented as Mean $\pm$ SD values

- **Mean flow pore size**

A capillary flow porometer (POROLUXTM1000, Germany) was used to calculate the mean flow pore sizes of the respective PLA10 and PLA10LEM10 electrospun matrices. The samples were wetted using n-Decane liquid have a low surface tension and pressed in a sealed chamber with nitrogen gas purging.

- ***In vitro* swelling studies**

A 10-day *in vitro* swelling study of the PLA10 and PLA10LEM10 electrospun scaffolds (n=3) was carried out by immersing the samples in PBS and incubating them at 37⁰C. The respective swelling ratio at each time point (Day 1, 4 and 10) was calculated using the formula:

71 $SR = W_W - W_D / W_D$ (Eqn 1)

72 Where W_W corresponds to the wet weight of the electrospun scaffold at a specific time point, W_D
73 represents the dry weight of the matrices before immersing in PBS whereas SR represents the
74 swelling ratio.

75 • **Differential scanning calorimetry (DSC)**

76 The thermal properties of the electrospun scaffolds were analyzed using DSC (Q2000, TA
77 Instruments). The samples post weighing were subjected to a heating and a cooling cycle from
78 30°C to 220°C and vice-versa at a scanning rate of 10°C/min. The corresponding glass transition
79 temperature (T_g), fusion temperature (T_f) and temperature of crystallization (T_c) were evaluated
80 from the respective thermograms. The % crystallinity of the samples was further calculated from
81 the formula:

82 $\% \text{ Crystallinity } (\chi) = \Delta H_f / \Delta H_f^0$ (Eqn 2)

83 Where the enthalpy of fusion (ΔH_f^0) of 100 % crystalline PLA is 93.7 J/g.

84 • **X-ray diffraction**

85 X-ray diffraction was carried out using an XPERT-PRO diffractometer to measure the crystallinity
86 of the PLA10 and PLA10LEM10 samples. A scanning 2θ range of 5-90° with a step size of 0.02°
87 was used for the analysis. The target used was a Cu K α target that was set at a voltage of 45 kV
88 and current of 40 mA⁴.

89 • **Energy-dispersive X-ray spectroscopy (EDX)**

90 EDX spectroscopy was carried out using the instrument OXFORD INSTRUMENTS XMAX to
91 evaluate the incorporation of the LEM into the PLA polymer. An accelerating voltage of 10 kV
92 and a spot size of 3 μ m were used for the analysis. The presence of carbon (C), hydrogen (H) and
93 nitrogen (N) was assessed using INCA software⁵.

94 • **Attenuated total reflectance-fourier transform infra-red spectroscopy (ATR-FTIR)**

95 The incorporation of new functional groups with the addition of LEM into the PLA was confirmed
96 using FTIR. The respective spectra of the given samples (PLA10 and PLA10LEM10 were
97 obtained in Attenuated Total Reflectance (ATR) mode using a NICOLET - IS-50 FTIR
98 Spectrometer (Thermo Fisher Scientific. The respective spectra were plotted using OriginPro 2021
99 (Origin Lab Corporation, Northampton, MA, USA)

100 • **Contact Angle**

101 The contact angle measurements were carried out using Drop Shape Analyzer instrument
102 (DSA100E) to evaluate the hydrophobicity/hydrophilicity of the electrospun PLA10 and
103 PLA10LEM10 mats. A 2 μ L deionized water droplet was dropped at three different positions and
104 the corresponding mean contact angle was evaluated.

105 • **Tensile property**

106 A Tinius Olsen universal testing machine was used to analyze the mechanical properties of the
107 electrospun matrices. Briefly, the PLA10 and PLA10LEM10 matrices were prepared in equal
108 dimensions (4 cm x 1 cm) and their respective thickness was evaluated using a thickness gauze.
109 The tensile strength was calculated using a load of 10 N and a stretching speed of 10 mm/min. The
110 corresponding tensile modulus was evaluated from the slope of the stress-strain curve⁶.

111 **5. Sterilization of the scaffolds for *in vitro* culture**

112 The electrospun nanofiber scaffolds were allowed for drying overnight in a vacuum hood. Later
113 the scaffolds were washed with 70 % ethanol thrice and sterilized with UV light for one hour. Then
114 the scaffolds were washed with PBS (Himedia Laboratories) thrice for 5 min each and soaked in
115 complete DMEM (10% FBS and 1% anti-biotic) (Gibco) overnight to make the surface of the
116 scaffolds suitable for the attachment of the hepatocytes.

6. Isolation of primary rat hepatocytes and characterization

Experimental animals used for the isolation of hepatocytes were 8-week-old Sprague Dawley rats, female, n=5, weighing around 300-350g as previously reported¹. The viability of the isolated hepatocytes was assessed with Trypan Blue staining (Himedia laboratories) and was also characterized with immunostaining for Albumin (Santacruz) after 24 hrs. of plating. The isolated primary rat hepatocytes (>90% viable) were seeded onto the sterilized scaffolds at a density of 10^5 cells / well, in a 24-well plate (Corning).

7. Characterization of the isolated primary hepatocytes

The isolated primary hepatocytes were characterized by phenotypic Albumin (Santacruz laboratories) expression at Day1 after seeding the cells on rat tail collagen-coated plates (RTC) (Gibco). The cells were washed with PBS (1x - pH 7.4) (Himedia) thrice 5 min each and permeabilized with Triton-X-100 (0.1% in PBS) (Himedia) for 15 min, followed by three PBS washes. They were then blocked with BSA (2% in PBS) (Himedia) for 1 hour, and incubated with the primary antibody, Albumin (Santacruz Biotechnology) (1:500 in 2% BSA) overnight. Later the cells were washed with PBS thrice and incubated with the secondary antibody Anti-mouse (Sigma-Aldrich) (1:1000 in 2% BSA) for 2 hours. Then incubation with DAPI (1:2000 in 2% BSA) for 5 min, followed by PBS wash to remove the background. The cells were then viewed under a confocal microscope (Leica LAS X systems)

8. Viability and functionality assessment of the cultured hepatocytes

The cells were washed with PBS (1x - pH 7.4) thrice for 5 min each and were incubated with Calcein (1 μ M) (Thermofisher scientific) and Sytox (1 μ M) (Thermofisher scientific) for 30 min followed by PBS washes, thrice. The cells were then viewed under a confocal microscope (Leica LAS X confocal microscope) at 494nm and 547nm wavelength for Calcein and Sytox respectively.

The cells were also assessed for albumin expression via Immunostaining on day1, 4 and 10. The morphology of the cultured hepatocytes was assessed via actin filament staining with Flash Phalloidin™ Red 594 (Biolegend). The secreted albumin of the cultured primary rat hepatocytes on conventional RTC plates and fabricated nano-fiber scaffolds was assessed from the conditioned media with Rat-Albumin specific ELISA kit (E-EL-R0362) at days 1,4 and 10 respectively.

9. Gene expression analysis of the cultured hepatocytes

Gene expression analysis of the hepatocytes cultured in the collagen-coated plates and nanofiber scaffolds was performed as mentioned earlier². Briefly, RNA was isolated using Trizol-chloroform (Thermo scientific) method and cDNA was prepared according to the kit protocol (Thermo Scientific™ RevertAid First Strand cDNA Synthesis Kit). Relative gene expression of Hepatocyte specific markers such as ASGR1, HNF4A and Albumin was assessed with the q-PCR technique (CFX96 Touch Real-Time PCR Detection System-BIORAD Systems). The primers used in this study are given in Supplementary Table 7.

10. CYP enzyme activity estimation

Cytochrome P450 (CYP) 1A2 enzyme activity was measured by incubating the cultured hepatocytes with Phenobarbital (potent inducer of the CYP1A2 enzyme) at varying concentrations such as 1mM, 1.5mM, 2mM for 4hrs, the cells were then washed with PBS 1x and the luciferin conjugate (Promega Life Sciences, Madison, WI) was added and incubated at 37° C for 30 minutes, later the detection reagent (Promega Life Sciences, Madison, WI) was added and kept in shaking for 10mins, as per the manufacturer's recommendations luminescence was quantified with Synergy multimode reader

11. Drug toxicity assessment

Isolated primary rat hepatocytes cultured on conventional Rat Tail Collagen (RTC) plates and on nanofiber scaffolds were washed with 1X PBS thrice and treated with the drugs for 24 hrs. After 24 hours, the cells were washed with 1X PBS twice and incubated with CellTiter-Glo® Substrate (CellTiter-Glo® Luminescent Cell Viability Assay kit- Promega) for 10 minutes at room temperature in a shaker at 37° C. The luminescence was recorded and plotted according to the drug concentration after subtracting the obtained values from the background luminescence.

Electrospinning parameters

Flow rate	1.5ml/h
Voltage	15kV
Distance	12cm
Needle size	24G
Humidity	30 to 40%
Viscosity	10% solution

Table S1- Standardized electrospinning parameters for fabrication of nanofiber scaffolds with LEM suitable for hepatic cells

Type	Polymer concentration (w% /v)	LEM concentration (w% /v)	Mixing ratio
PLA5	5% PLA	-	-
PLA10	10% PLA	-	-
PLA10 LEM5	10% PLA	5% LEM	1:1
PLA10 LEM10	10% PLA	10% LEM	1:1
PLA10 LEM 20	10% PLA	20% LEM	1:1

Table S2- PLA and Liver extracellular matrix mixed at different ratios and concentrations and utilized for electrospinning.

Drug	Concentration used
Acetaminophen	5 mM
	10 mM
	15 mM
	20 mM
Rifampicin	25 μ M
	50 μ M
	75 μ M
	100 μ M
	150 μ M

Table S3- Drug concentrations used for hepatotoxic screening

Sample ID	Carbon (%)	Oxygen (%)	Nitrogen (%)
PLA10	69.23 $\pm$ 0.56	30.77 $\pm$ 3.0	-
PLA10 LEM5	57.90 $\pm$ 2.45	32.78 $\pm$ 7.12	5.90 $\pm$ 3.27
PLA10 LEM10	60.88 $\pm$ 2.5	28.24 $\pm$ 4.56	10.88 $\pm$ 1.25
PLA10 LEM20	55.69 $\pm$ 1.49	30.9 $\pm$ 4.18	19.23 $\pm$ 4.67

Table S4- Elemental analysis of the fabricated scaffolds by EDX Analysis

Sample ID	Tensile modulus (kPa)	Tensile strength (kPa)	Elongation at break (%)	Viscosity (mPas)
PLA10	1285.5±332.5	11.2584±8.6	162.48±5.3	181.46±93.2
PLA10 LEM10	422.3±462.6	2.4484±.662	18.68±3.2	34.578±62.3

Table S5 - Physical characterization of the fabricated scaffolds

Sample ID	Glass transition temperature (T _g °C)	Fusion temperature (T _f °C)	Crystallization temperature (T _c °C)	Enthalpy of fusion	% crystallinity
PLA10	62.12±2.5	142.04±8.9	86.02±4.5	25.13±6.2	26.80±4.3
PLA10LEM10	63.37±4.75	140.97±1.3	85.66±.32	30.06±7.6	32.08±0.5

Table S6 - Thermal characteristics and crystallinity of the electrospun PLA10 and PLA10 LEM10

nano-fiber scaffolds

Primer	Sequence	Company
ASGR1	CACTCCACGCTGCTCTTC CTGACAGGATTTTCAGCAC	Eurofins Scientific
HNF4A	TGAAGGTGCCAACCTCAACTC GACTCACACACATCCGTGATG	Eurofins Scientific
ALBUMIN	TTCAGTCCCTCCTTACAGCC TCCAAGGCAGAATACGGTGAC	Eurofins Scientific

Table S7- Primer sequences used in the q-PCR Analysis

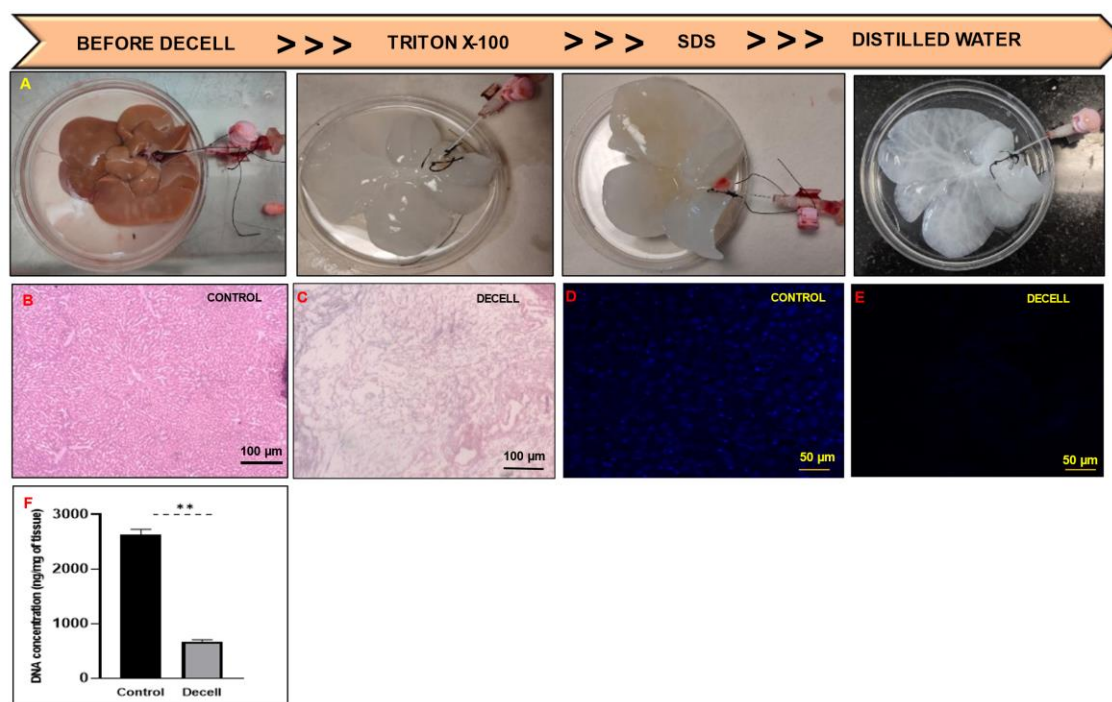


Fig S1- Characterization of the decell liver tissues in terms of cellular content. Healthy control SD rat liver decellularized via portal vein (A) Appearance of the liver throughout the decellularization process. (B) H&E staining shows the presence of the nucleus before decell and (C) the absence of the same after the decellularization process. (D) Nuclear staining of the control liver tissues with DAPI (E) Nuclear staining of the decell liver tissues with DAPI (F) DNA estimation of control and decellularized liver sample, confirming the removal of cells post-decellularization process. Statistical analysis was performed with GraphPad Prism, t-test in comparison with the control group (**P<0.01)

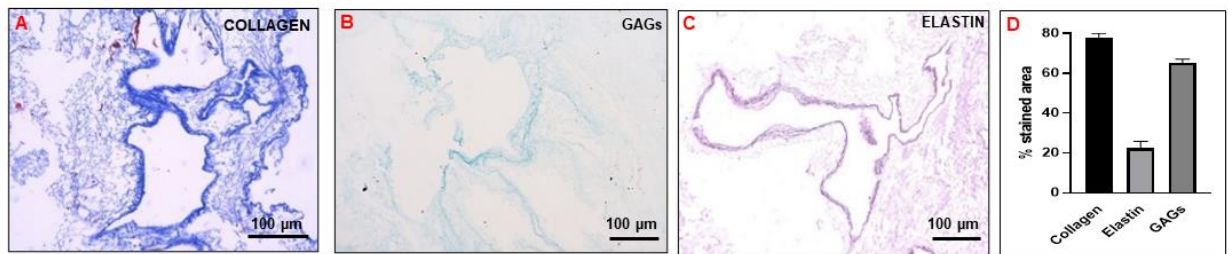


Fig S2- Characterization of decell liver tissues in terms of ECM protein content (A) Masson Trichrome staining showing the presence of collagen intact post-decell (B) Alcian blue staining shows the presence of glycosaminoglycans intact post-decellularization. (C) Orcein staining shows the presence of elastin fibers in the extracellular matrix post-decellularization (D) Quantification of matrix proteins in the decell tissues performed with ImageJ.

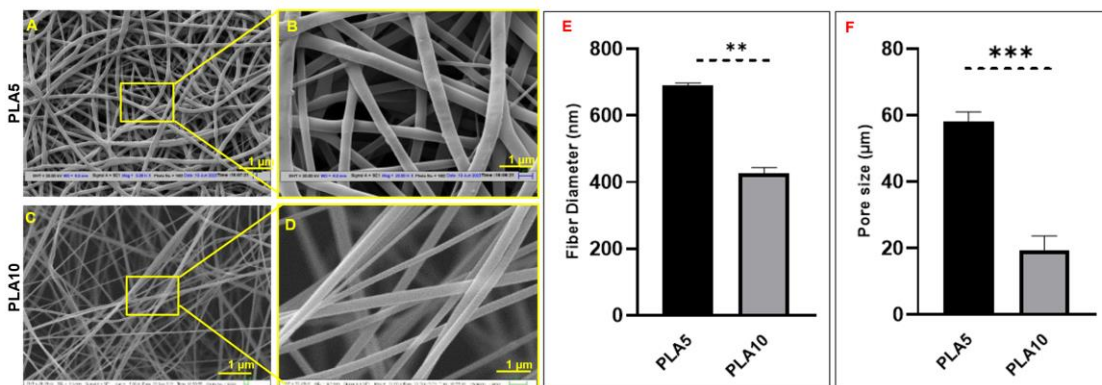
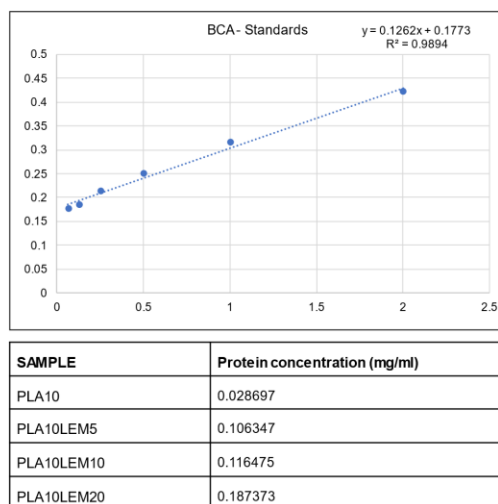


Fig S3- Physical characterization of PLA scaffolds. (A) SEM images of Electrospun PLA5 (B) PLA 5 scaffolds at 2000x magnification (C) SEM images of Electrospun PLA10 (D) PLA10 scaffolds at 2000x magnification (E) Measurement of fiber diameter from the SEM images. (F)

204 Measurement of Pore size from the SEM images. Quantification performed with ImageJ. Graphs
 205 were plotted and statistical analysis performed with Graphpad Prism. t-test between PLA5 vs
 206 PLA10 (**P<0.05; ***P<0.001)



207

208 **Fig S4-** Protein estimation of the fabricated nanofiber scaffolds

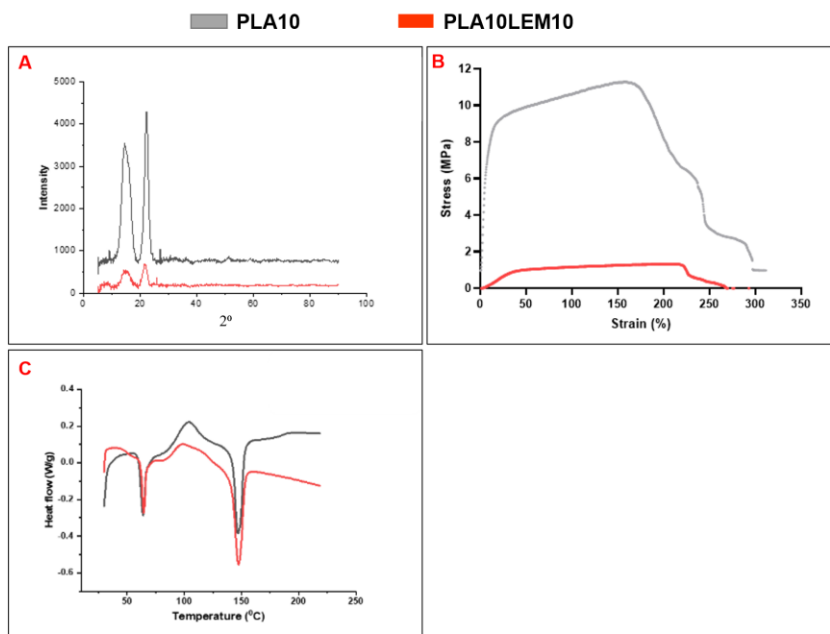


Fig S5 – Physical characterization of the PLA10LEM10 Scaffolds **A.** X-ray diffraction analysis of PLA10 and PLA10LEM10 scaffolds, **B.** Stress-strain curve of the PLA10 and PLA10LEM10 scaffolds **C.** DSC analysis of PLA10 and PLA10LEM10.

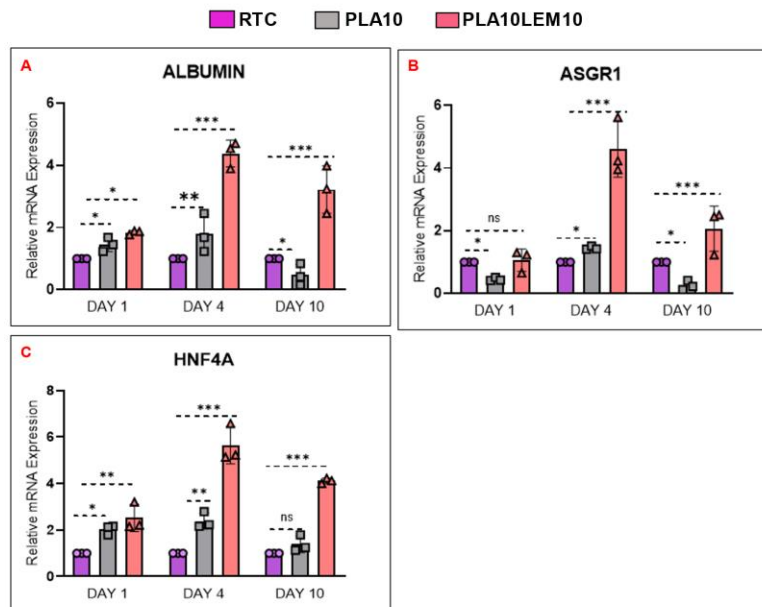


Fig S6- Gene expression analysis of the cultured primary hepatocytes on conventional RTC plates and fabricated nanofiber scaffolds. A. Albumin Gene expression. B. ASGR1 Gene expression. C. HNF4A Gene expression. Statistical analysis was performed with GraphPad prism T-test with Mann-Whitney test P value determined with respect to RTC . (*P<0.05; ** P<0.001;* P<0.0001)**

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